Why are complex-valued signals preferred in communication system analysis if physical signals are real?
Under what condition is the sampling of continuous signals considered lossless?
What type of noise is introduced into a communication system during the quantization process?
Why is channel noise generally assumed to follow a Gaussian distribution?
If a signal $y(t)$ is the result of convolving two signals with bandwidths B_1 and B_2, what is the upper bound of its bandwidth?
Formula: Energy Spectral Density (ESD)
What is the primary goal of employing multiplexing techniques like TDMA or FDMA?

Term: CDMA

According to the Uncertainty Principle, why can we never achieve perfectly band-limited signals?
How is 90% bandwidth defined in the frequency domain?
What characterizes the '3dB bandwidth' of a signal or system?
Formula: Decibel (dB) power ratio
What is the purpose of the Paley-Wiener Criterion?
Formula: Standard Communication System Model
In real-life hardware, what specific type of phase response is achievable for generated filters?
Formula: Simple Wired Channel Model
What phenomenon occurs in wired communication due to signal reflections from objects?
Why are GHz frequencies specifically used for modern wireless transmission?
In modulation, the baseband signal $M(\omega)$ is converted to a higher frequency f_c through a process
How is a real signal $x(t)$ represented using its complex baseband components $m_{Re}(t)$ and $m_{Im}(t)$?

Term: Passband Signal
In an up-conversion circuit, what component generates the high-frequency carrier wave?

What happens to the recovered signal if the receiver's Local Oscillator is not synchronized with the transmitter?
What is the maximum theoretical efficiency (η) for a standard Double Sideband Amplitude Modulation

Formula: AM Modulation Index (α)
How does High Peak-to-Average Power Ratio (PAPR) generally affect communication system performance?
What mathematical operator is used to generate the quadrature component needed for Single Sideband (SSB)
Formula: Hilbert Transform frequency response
In Double Sideband Suppressed Carrier (DSB-SC), what happens to the signal at the origin ($f=0$) in the

Concept: QAM (Quadrature Amplitude Modulation)

In Angle Modulation, how is Frequency Modulation (FM) distinguished from Phase Modulation (PM)?
What receiver architecture translates the carrier frequency to a fixed Intermediate Frequency (IF)?

Term: Image Frequency

Formula: Carson's Rule for FM Bandwidth
What hardware component is typically used to demodulate FM signals by tracking phase?

Under what condition is a random process $X(t)$ considered Wide Sense Stationary (WSS)?

What does the Wiener-Khinchin Theorem establish regarding a WSS process?

Concept: Ergodicity
What is the defining characteristic of a Cyclostationary process?
Formula: Power Spectral Density (PSD) of AWGN
What is the primary objective of a Matched Filter in a receiver?
Formula: Impulse response of a Matched Filter for a signal $s(t)$

Term: Noise Figure (F)
What formula is used to calculate the total noise figure of cascaded amplifier stages?
In quantization, how does the Signal-to-Quantization-Noise Ratio (SQNR) change for every additional bit added to the encoder?

What is the purpose of the Lloyd-Max Optimizer in communication systems?

Term: Companding

Why is Differential PCM (DPCM) more efficient than standard PCM?
What error occurs in Delta Modulation when the signal's slope exceeds the maximum possible step rate?
What is the Nyquist Rate requirement for sampling a signal with bandwidth B?
In digital communication, what filter is used to prevent frequency aliasing before sampling?
Formula: Propagation loss in free space
How does an increase in carrier frequency f_c impact the required size of a transmitter antenna?
What is the spectral shape of an 'impulse' in the frequency domain?
In PLL analysis, what happens if the input message has a significant DC component?

Term: BPSK (Binary Phase Shift Keying)
What does a 'Circular Symmetric Complex Gaussian' distribution imply about the real and imaginary noise

Concept: Bandpass Sampling
Formula: Signal-to-Noise Ratio (SNR)
What is the effect of 'Spectral Inversion' during down-conversion?
In a Superheterodyne receiver, what is the role of the RF filter?
What type of modulation uses the formula $s(t) = A_c [1 + \alpha m(t)] \cos(2\pi f_c t)$?
Why is Narrowband FM (NBFM) considered similar to AM?
What is the function of a 'Vocoder' or 'VCO' in FM modulation?
In signal space analysis, what must be true for two signals $\psi_i(t)$ and $\psi_j(t)$ to be an orthonormal basis?

What is the 'Link Budget' in a communication system design?

In the context of quantization, what is the 'Dynamic Range'?
Which standard companding law is widely used in European digital telephony?
What is the primary disadvantage of using a Homodyne (Zero-IF) receiver?
In random process theory, what is the relation between the cross-correlation $R_{xy}(\tau)$ and $R_{yx}(\tau)$?

They simplify mathematical analysis by providing a complex representation ($x(t) \rightarrow X(\omega)$).
The signal must be bandlimited.
Random additive noise.
It is a consequence of the Central Limit Theorem.

$y(t) \le \min(B_1, B_2)$.
$|X(f)|^2$
To ensure that transmitted signals are orthogonal to prevent interference.
Definition: Code Division Multiplexing, which utilizes phase differences (e.g., $\sin$ and $\cos$) for separation.
A signal cannot be limited in both the time domain and the frequency domain simultaneously.
The range $[-a, a]$ that contains 90% of the total energy spectral density.
The frequency range where the power drops to half of its peak value.
$10 \log_{10}(\frac{P_1}{P_2})$
To check if a specific filter transfer function is physically realizable.
$r(t) = h(t) * m(t) + n(t)$ where $r(t)$ is the receiver signal and $n(t)$ is AWGN.
Linear phase.
$h(t) = a_0 \delta(t - t_0)$ representing attenuation and delay.
Inter-Symbol Interference (ISI).
To satisfy antenna length constraints, which decrease as frequency increases.
Up-conversion
$x(t) = m_{Re}(t) \cos(2\pi f_c t) - m_{Im}(t) \sin(2\pi f_c t)$.
Definition: A signal centered around a high carrier frequency that does not contain a DC component.
Local Oscillator (LO).
A phase difference occurs, leading to a loss of orthogonality between components.
50%.
$\alpha = \frac{M}{A_c}$ where M is the peak message value and A_c is carrier amplitude.
It reduces power efficiency and is considered undesirable.
The Hilbert Transform.
$H(f) = -j \text{sgn}(f)$
The carrier component is removed, leaving only the sidebands.
Definition: A technique using two independent message signals modulated onto orthogonal carriers (sine and cosine).
In FM, the message signal controls the derivative of the phase (frequency deviation).
Superheterodyne receiver.
Definition: An unwanted frequency at $f_c \pm 2f_{IF}$ that can interfere with the desired signal in heterodyne receivers.
$BW \approx 2(\Delta f + f_m)$ where Δf is frequency deviation and f_m is message bandwidth.
Phase Locked Loop (PLL).

Its mean is constant and its auto-correlation depends only on the time difference τ.

The Power Spectral Density (PSD) is the Fourier Transform of the auto-correlation function.

Definition: A property where the time average of a single realization equals the ensemble average of the process.

Its statistics (like mean and auto-correlation) vary periodically with time.

$S_n(f) = \frac{N_0}{2}$ (constant across all frequencies).

To maximize the Signal-to-Noise Ratio (SNR) at the sampling instant.

$h(t) = s^*(T - t)$

Definition: The ratio of the input SNR to the output SNR of a device, always ≥ 1.

Friis Formula.

It increases by approximately 6 dB.

To determine the optimal non-uniform quantization levels for a specific signal PDF.

Definition: A process of compressing a signal before quantization and expanding it after to improve SQNR for non-uniform distributions.

It quantizes the difference between successive correlated samples, which has lower power than the raw signal.

Slope Overload.

$F_s > 2B$.

Anti-aliasing filter.

$L = (\frac{4\pi d}{\lambda})^2$

It allows for a smaller antenna size.

A flat spectrum containing all frequencies.

The integrator in the loop can cause the system to become unstable.

Definition: A digital modulation scheme where data is transmitted by shifting the phase of the carrier by 180 degrees.

They are independent, identically distributed (IID) Gaussian variables.

Definition: A technique to sample a signal centered at f_c with a rate lower than $2f_{max}$, provided certain overlap conditions are met.

$\frac{P_{signal}}{P_{noise}}$

The sidebands of the signal spectrum are flipped.

To reject the image frequency before it reaches the mixer.

Double Sideband Large Carrier (DSB-LC) or Standard AM.

Its bandwidth is approximately $2B$ and it contains a large carrier component.

It translates a voltage signal directly into a corresponding frequency shift.

They must be orthogonal and each have unit energy.

An accounting of all gains and losses from the transmitter to the receiver to ensure a target SNR.

The total range of amplitudes $[-x_{max}, x_{max}]$ that the quantizer can process.

A-law.

It is highly sensitive to DC offsets and $1/f$ noise.

$R_{xy}(\tau) = R_{yx}^*(-\tau)$.